

DATA 604 – Week 3 Assignment

Queue Modeling and Simulation: Laboratory Sample Processing

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1. Narrative of the Process and Attributes Considered

For this assignment, I modeled a routine laboratory sample-processing system. Laboratory samples arrive throughout the workday and must pass through several steps before processing is complete. Because each step has limited resources, samples may have to wait in a queue when the required staff member or equipment is already in use.

Each laboratory sample represents an entity in the simulation. The resources include registration staff, a laboratory technician, an analyzer, and a scientist responsible for reviewing the results.

The process consists of four main stages:

Process Step	Estimated Processing Time	Resource Capacity
Registration	3 minutes	1 staff member
Sample Preparation	8 minutes	1 technician
Testing/Analysis	15 minutes	1 analyzer
Result Review	5 minutes	1 scientist

Samples were modeled as arriving approximately every 5 to 10 minutes. The simulation represented an eight-hour workday, or 480 minutes.

When a sample arrives, it first goes through registration. It then moves to sample preparation, followed by testing and analysis. Finally, the result is reviewed before the sample is considered completed. If the required resource is unavailable at any stage, the sample waits until that resource becomes available.

2. Analysis of Bottlenecks in the Queue

The original simulation showed that the Testing/Analysis stage was the primary bottleneck.

The average waiting times from the original process were:

Process Step	Average Waiting Time
Registration	0.00 minutes
Preparation	22.98 minutes
Testing/Analysis	108.31 minutes
Result Review	0.00 minutes

Testing had an average waiting time of 108.31 minutes, which was substantially greater than the waiting times at the other stages.

The reason for this bottleneck is that testing takes approximately 15 minutes per sample, while new samples arrive every 5 to 10 minutes. With only one analyzer available, samples can arrive faster than the analyzer can process them. As a result, samples accumulate while waiting for testing.

Preparation also showed an average waiting time of 22.98 minutes, but the testing delay was considerably larger.

In the original eight-hour simulation, 30 samples were completed, and completed samples spent an average of 146.70 minutes in the system.

3. Suggested Changes

To improve the process, I increased the capacity of the Testing/Analysis stage from one analyzer to two analyzers.

The testing time remained at 15 minutes per sample, but the additional analyzer allowed two samples to undergo testing simultaneously.

I kept the other resource capacities and processing times unchanged. This allowed the modified simulation to specifically measure the effect of increasing testing capacity rather than changing several parts of the process at the same time.

The expectation was that adding testing capacity would reduce the long testing wait, decrease the total amount of time samples spent in the system, and allow more samples to be completed during the workday.

4. Comparison of Results

The modified simulation produced a substantial improvement compared with the original process.

Measure	Original Process	Modified Process
Samples Completed	30	57
Registration Wait	0.00 min	0.00 min
Preparation Wait	22.98 min	22.98 min
Testing Wait	108.31 min	0.00 min
Review Wait	0.00 min	0.00 min
Average Total Time in System	146.70 min	52.94 min

Increasing the number of analyzers from one to two reduced the observed average testing wait from 108.31 minutes to 0.00 minutes in this simulation run.

The average total time spent in the system decreased from 146.70 minutes to 52.94 minutes, representing a 63.91% reduction.

The number of completed samples increased from 30 to 57, representing a 90.00% increase in completed samples during the simulated eight-hour workday.

Although the testing bottleneck was resolved in this simulation run, sample preparation continued to have an average waiting time of 22.98 minutes. This indicates that preparation becomes the next constraint that could be investigated if additional improvements were desired.

Conclusion

This simulation demonstrated how queue modeling can be used to identify bottlenecks and evaluate potential process improvements. In the original laboratory process, Testing/Analysis created the greatest delay because a single analyzer could not process samples as quickly as they arrived.

Increasing the number of analyzers from one to two eliminated the observed testing wait in this simulation run and substantially reduced the average total time completed samples spent in the system. It also increased the number of samples completed during the simulated workday.

The comparison demonstrates the usefulness of simulation for testing changes before implementing them in an actual process. It also demonstrates that resolving one bottleneck can reveal another constraint, as sample preparation became the stage with the greatest average waiting time after testing capacity was increased.